Evaluation of validation strategies and transferability of EEG attention decoding: a comparative analysis on open datasets

Sanzhar Sovet[1], Beibit Abdikenov[1*], Manzura Zholdasova[2,3], Altyngul Kamzanova[2], Medet Mukushev[1], Almira Kustubayeva[2,3]

1 Science and Innovation Center "Artificial Intelligence", Astana IT University, Astana, Kazakhstan

2 Department of Biophysics, Biomedicine, and Neuroscience, Al-Farabi Kazakh National University, Almaty, Kazakhstan

3 Brain Institute, Al-Farabi Kazakh National University, Almaty, Kazakhstan

* beibit.abdikenov@astanait.edu.kz

Abstract

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Introduction

Electroencephalography (EEG) is a technique for recording brain activity using multichannel scalp electrodes that measure voltage signals reflecting underlying cognitive processes. It is widely used in neuroscience and brain–computer interface (BCI) research due to its non-invasive nature, high temporal resolution, portability and relatively low cost [1]. At the same time, EEG signals are inherently noisy, non-stationary, and exhibit high inter-subject and inter-session variability [2]. In addition, these challenges are compounded by the scarcity of large, diverse, and well-annotated open EEG datasets [3]. Consequently, developing a robust decoding model with strong generalization capabilities across subjects, sessions and tasks remains an open and central challenge [2, 4]. One promising approach to solve these problems is transfer learning (TL). TL reuses representations already learned on one or more source problems to reduce the amount of labeled data and training effort a related target problem would otherwise require [5]. Its success depends not only on modeling and learning strategies but also on the degree of domain and label compatibility between source and target datasets.

For cognitive constructs such as attention, this compatibility is especially consequential and difficult to guarantee because attention has always been an ambiguous concept and is not a unitary construct with a single universally accepted operational definition [3, 6]. Rather, attention is an umbrella term encompassing a set of partially independent processes and functions, such as selecting relevant information, maintaining focus over time, shifting attention between tasks, and allocating cognitive resources. These processes may operate on both external sensory input and internally represented information, including information maintained in working memory [6–8]. The relative contributions of these processes may vary with task demands, sensory modalities, task goals, experimental instructions, assigned conditions, self-reported assessments, and other aspects of experimental design and label definition. Consequently, experimental paradigms described as measuring "attention" may probe non-equivalent configurations of cognitive processes. As a result, different datasets may represent different attention-related cognitive states while assigning them the same nominal labels.

Previous EEG transfer-learning research has been predominantly concerned with cross-subject and cross-session generalization, whereas cross-task, cross-dataset, and cross-paradigm transfer have received comparatively less systematic attention [2, 9, 10]. Consequently, systematic empirical evidence for source–target suitability across heterogeneous attention-related EEG datasets and tasks remains limited. Source–target suitability is often motivated conceptually or inferred from nominally aligned labels. This leaves unclear which source–target combinations provide useful zero-shot transfer, how much target discriminability is retained relative to target-specific baselines, and whether transferability is direction-dependent. Establishing such pre-adaptation baselines is necessary to justify source selection and to interpret gains or negative transfer from subsequent transfer-learning and domain-adaptation methods [11].

[IN PROGRESS — This paragraph is retained provisionally. Remove it if suitable experiments comparing operational label mappings cannot be completed.]

Here, we contribute to addressing this gap by systematically evaluating zero-shot generalization across two technically compatible public EEG datasets spanning multiple tasks and paradigms. We examine both within-dataset baseline performance and transfer across subjects, sessions, trials, tasks, and datasets under a common preprocessing pipeline, compare directed source–target pairs against target-specific baselines, and assess transferability using five decoders ranging from linear and tree-based baselines to deep neural networks. Together, these analyses yield reproducible pre-adaptation benchmarks for evaluating candidate sources and contextualizing both gains and negative transfer produced by subsequent transfer-learning and domain-adaptation methods.

Open-access datasets

EEG data for mental attention state detection (Dataset A)

The dataset described in [12] comprises **34 EEG recordings** collected from **five participants** over the course of **seven days**, with the fifth participant completing only **six sessions**. During each session, participants operated a virtual train using the *Microsoft Train Simulator*. Each recording lasted approximately **35 to 55 minutes**. All experiments in this study were performed with healthy volunteer participants chosen among students.

To ensure consistency, all sessions were conducted on the same predefined route, designed to require minimal interaction. This setup constitutes a **low-intensity control task**, minimizing variability due to task demands and aligning with *naturalistic stimulus-based protocols* as described in [13]. Each session was divided into **three sequential phases**, each designed to reflect a different target mental state: ***focused***, ***unfocused***, and ***drowsy***. The participants simulated these mental states based on standardized instructions provided by the experiment supervisor:

- **Focused (0–10 min):** Participants actively monitored and mentally engaged with the simulator, maintaining sustained attention to the virtual controls and screen events.
- **Unfocused (10–20 min):** Participants disengaged from the task, ceasing interaction and intentionally

avoiding attentional focus, while being instructed to keep their eyes open and remain awake.

- **Drowsy (20–30 min):** Participants were allowed to relax, close their eyes, and enter a drowsy or lightly dozing state, simulating reduced cognitive arousal.

EEG signals were recorded using a **modified Emotiv headset**, equipped with wet electrodes and supporting wireless Bluetooth transmission. The device recorded data at a **sampling rate of 128 Hz** and with a **voltage resolution of 0.51 μV**. The dataset includes **14 EEG channels**, labeled according to the 10–20 international system: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4.

SAM-40: dataset of 40-subject EEG recordings (Dataset B)

The **SAM-40** dataset, described in [14], contains EEG recordings from 40 healthy participants (14 females and 26 males) with a mean age of 21.5 years. The purpose of data collection was to investigate short-term stress responses induced by the performance of various cognitive tasks, which correspond to a *continuous cognitive engagement* paradigm as described in [13]. The experimental protocol included four conditions:

- **Resting state** — EEG recording during relaxation while listening to music.
- **Stroop Color-Word Test** — participants were required to identify the color of the ink in which a word was printed, while ignoring the word's semantic meaning. The task included two types of stimuli: *congruent* (the color name matched the ink color) and *incongruent* (the color name and ink color differed). Each execution of the task included 11 stimuli.
- **Arithmetic problem solving** — participants mentally solved a mathematical problem and determined whether the presented answer was correct, responding with a "thumbs up" or "thumbs down" gesture. Each execution of the task included 6 arithmetic stimuli.
- **Mirror Image Recognition** — participants were shown images and asked to determine whether they were mirror-symmetric or asymmetric, responding with the same gestures. Each execution of the task included 8 images.

Each task lasted 25 seconds, with 5-second rest intervals and a 10-second instruction period preceding the next task. After the completion of each trial, participants rated the level of stress experienced for each task on a 10-point scale, where 1 indicated minimal stress and 10 indicated maximal stress. For each participant, three trials were recorded using different sets of stimuli.

EEG signals were recorded using an Emotiv Epoc Flex headset with 32 wet electrodes, labeled according to the international 10–20 system. Data were acquired at a sampling rate of 128 Hz and a voltage resolution of 0.51 μV. The 32 channels included in the dataset are: Cz, Fz, Fp1, F7, F3, FC1, C3, FC5, FT9, T7, CP5, CP1, P3, P7, PO9, O1, Pz, Oz, O2, PO10, P8, P4, CP2, CP6, T8, FT10, FC6, C4, FC2, F4, F8, and Fp2.

Methodology

The methodological design of this study was developed to evaluate the generalizability and transferability of EEG-based attention decoding across different recording conditions, participants, and task paradigms. To this end, we adopted two publicly available datasets that differ in experimental protocols but share compatible technical specifications, which allows for a controlled comparative analysis. The datasets were harmonized through a binary operational labeling scheme that distinguishes conditions of high protocol-defined task demand from conditions of rest or task disengagement, enabling consistent evaluation across scenarios. This section outlines the rationale for dataset alignment, the strategies employed for cross-validation, and the experimental settings designed to assess within-dataset performance as well as cross-subject, cross-session, cross-task, and cross-dataset generalization.

All experiments were executed through a version-controlled computational workflow. For every run, the workflow stores the resolved configuration of data preparation, validation and model settings together with the run outputs needed to reconstruct the reported evaluation metrics. These records provide run-level provenance for the results and permit the correspondence between a reported metric and the experiment that produced it to be inspected. The complete implementation, experiment configurations and run-level artifacts are available at `https://github.com/sovetskiysn/eeg-validation-strategies`.

Dataset alignment and integration rationale

Effective evaluation of cross-task and cross-dataset generalization relies on the availability of datasets that are compatible both technically and conceptually. In this study, we integrate two publicly available EEG datasets that differ in their original purpose and labeling schemes, yet retain sufficient technical and conceptual compatibility for a meaningful comparison.

A major strength of this dataset combination lies in their acquisition setup. Both datasets were recorded using Emotiv EEG systems with the same sampling rate (128 Hz) and voltage resolution, although their original electrode montages differed (14 versus 32 channels). For cross-dataset analyses, only the 12 channels common to both datasets were retained. Differences in EEG recording systems and electrode configurations can hinder cross-dataset transferability [15]. Restricting the analysis to the shared channels substantially reduces this configuration-related mismatch, although it does not eliminate all acquisition-related distribution shifts. Within the limited set of publicly available datasets suitable for this comparison, this pair offered a comparatively compatible basis for the present comparative evaluation.

The conceptual rationale for combining the datasets rests on the cognitive conditions represented in their respective protocols. The mental attention-state dataset contains explicit attention-state labels (*Focused*, *Unfocused*, and *Drowsy*) collected during a low-interaction simulation task, whereas SAM-40 was originally designed for stress assessment. Both capture cognitive conditions strongly tied to attentional processes. The SAM-40 dataset includes tasks widely recognized in cognitive neuroscience for eliciting high levels of attentional engagement:

- **Stroop Color Test** — probes both *selective* and *sustained attention*, requiring participants to suppress automatic reading responses and continually focus on the ink color throughout the task [16–18].
- **Mental arithmetic task** — demands *sustained internal attention* and working memory for continuous manipulation of numerical information [19, 20].
- **Mirror image recognition task** — engages visuospatial processing and requires both *selective* and *sustained attention* to discriminate between symmetric and asymmetric patterns, relying on mental rotation processes that demand continuous cognitive effort [21–23].

The simulator-based and task-based datasets correspond, respectively, to the *naturalistic stimulus-based protocols* and *continuous cognitive engagement* paradigms described in [13]. In the unified labeling scheme, *Rest* and *Unfocused* are treated as low-attention states, whereas *Focused*, *Stroop*, *Arithmetic*, and *Mirror recognition* are treated as high-attention states. The *Drowsy* phase is excluded from this mapping and from all subsequent preprocessing and analyses because permitted eye closure and drowsiness may confound the comparison with differences in arousal, visual input, and ocular activity. This mapping is summarized in Table 1.

Table 1. Unified binary attention label mapping across datasets.

Attention State	Conditions
High Attention	Focused, Stroop, Arithmetic, Mirror
Low Attention	Unfocused, Rest

Validation strategies [revise need]

We evaluated the decoders under four validation protocols, each addressing a distinct generalization question. To preserve the independence of the training and test data, all windows originating from the same recording unit were assigned to the same partition. This prevents overlapping windows from being divided between training and test sets, reducing the risk of overly optimistic performance estimates due to temporal autocorrelation between neighboring EEG windows [13]. The within-domain baseline estimates performance when training and test data share the same dataset and condition composition but originate from different recording units. The remaining protocols examine whether discriminative performance is retained when training and testing are separated by participant, task composition, or dataset. These protocols were applied only where the structure of each release supported their interpretation: multiple tasks in SAM-40 enabled cross-task validation. Model fitting was confined to the training data, and the resulting decoder was evaluated on held-out data without target-specific adaptation. Any class imbalance was addressed through inverse-frequency class weighting calculated independently within each training fold, while test data retained their observed class distribution.

Baseline evaluation. Reference performance was estimated using five-fold stratified group cross-validation, separately for the mental attention-state dataset, the complete SAM-40 composition, and each task-specific composition of SAM-40. The grouping variable was the physical recording unit: one selected session in the mental attention-state dataset and one run-level recording unit in SAM-40, comprising all task recordings included in the analysis for that run. Training and test folds thus shared the dataset and condition composition and could contain data from the same participants, but never from the same recording unit. These estimates serve as within-domain comparators for the transfer protocols.

Cross-subject validation. In each leave-one-subject-out fold, the decoder was trained on all selected data from the remaining participants and tested on all selected data from the held-out participant. The resulting performance quantifies zero-shot generalization to an unseen participant.

Cross-task validation. In SAM-40, each task composition represented the same binary contrast of relaxation versus high-demand task conditions, rather than classification of task identity. Cross-task transfer therefore asked whether a decoder fitted to this contrast under one task composition retained discriminative performance under another.

The cross-task folds were also subject-disjoint. For each fold, the source composition from all but one participant formed the training set, whereas the target composition from the held-out participant formed the test set. The protocol thus evaluates the combined shift to an unseen participant and a different task composition, rather than within-participant transfer between tasks. This distinction prevents participant-specific EEG patterns from contributing to apparent cross-task generalization [24, 25].

Cross-dataset validation. To evaluate transfer between datasets, models were trained on all available data from a source composition and tested on a composition from the other dataset, in both transfer directions and without target-specific adaptation. SAM-40 was considered as a complete composition and through its task-specific compositions. Unlike the preceding protocols, cross-dataset validation changes several factors simultaneously, including the participant population, experimental paradigm, recording setup, and condition definitions. It therefore measures transfer under their compound distribution shift rather than attributing a performance change to any single factor [15]. The common sampling rate, harmonized binary labels, and shared 12-channel montage establish a compatible input and prediction space, but do not remove the substantive differences between the two datasets.

Data preparation [revise need]

The same preparation pipeline was applied to both datasets and across all validation protocols. Its filtering, ICA, channel alignment and windowing parameters were fixed in advance and were not tuned separately for a dataset, task composition, validation protocol or decoder. The purpose was to place recordings in a common input space while limiting preprocessing itself as an additional source of variation. Differences between validation results are therefore interpreted against a fixed preparation procedure.

Recording and label selection. Recordings were selected before signal processing. Following the source study, the first two recording sessions for each participant in the mental attention-state dataset were intended for habituation to the simulator and were not included in the analyses [12]. The *Drowsy* condition was excluded from the binary classification analysis because participants were permitted to close their eyes and enter a drowsy state. This condition differs from the *Focused* and *Unfocused* conditions not only in protocol-defined task demand, but also in arousal, visual input and ocular activity. Including it in the low-attention class would therefore confound the intended contrast. The drowsy segment was removed before filtering and ICA, so no drowsy epochs were generated or presented to a classifier. In contrast, the complete collection of SAM-40 recordings was retained for analysis. SAM-40 provides the task recordings of each trial as separate files. Within each scenario, the task recordings selected from one trial were grouped into a single recording unit, whereas task recordings not selected by the scenario were not carried forward to later preparation steps. Accordingly, the binary mapping in Table 1 was applied to the retained recordings from the two datasets.

Filtering and ICA cleaning. Filtering was applied to each recording, and ICA was fitted within participant-specific acquisition groupings. For the mental attention-state dataset, each grouping corresponded to a selected session; for SAM-40, it comprised the selected task recordings of a participant. To prevent preprocessing from becoming an additional source of variation, we used a deliberately restrained, conventional procedure: standard filtering followed by ICA-based artifact cleaning. Keeping preparation deliberately simple allows differences in performance to be interpreted primarily in relation to the recordings and validation protocols, rather than to a complex preparation pipeline. EEG channels were notch-filtered at 50 Hz, band-pass filtered from 1 to 45 Hz and rereferenced to the average reference. Artifact correction used independent component analysis with extended-Infomax optimization, retaining components that explained 99% of the variance. The resulting components were automatically classified from their spatial, spectral and temporal characteristics; components identified as eye-blink or muscle artifacts with a probability of at least 0.80 were removed. The ICA random state was fixed by the experiment-wide configuration.

Channel alignment and window segmentation. After signal cleaning, each recording was restricted to the 12 channels shared by the two datasets: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7 and T8. Each retained labelled block was segmented into 5-s windows with 0.5-s overlap. Every retained window inherited the label of its source block and metadata identifying its dataset, participant, session, task, run, recording unit and temporal onset. The recording-unit identifier was used by the validation protocols to ensure that all overlapping windows from a unit remained on the same side of a split.

The fixed selection, cleaning, alignment and segmentation steps therefore yielded a deterministic analytical window set for each dataset composition. All model-specific input representations were derived from these same prepared windows, rather than from separately prepared data. Table 2 summarizes the resulting compositions, including their retained recording units, window counts and observed class distributions. Class weights were calculated independently within each training fold. All Dataset B configurations contain the same 40 participants and differ only in the task recordings retained alongside the resting-state recordings.

Table 2. Analytical dataset compositions.

Dataset configuration	Retained conditions	Participants	Recording units	Windows, n (low / high)	Class distribution, % (low / high)
Full Dataset A	Focused, Unfocused	5	24	3,192 / 3,192	50.0 / 50.0
Full Dataset B	Rest, Stroop, Arithmetic, Mirror	40	120	600 / 1,800	25.0 / 75.0
Dataset B: Stroop	Rest, Stroop	40	120	600 / 600	50.0 / 50.0
Dataset B: Arithmetic	Rest, Arithmetic	40	120	600 / 600	50.0 / 50.0
Dataset B: Mirror	Rest, Mirror	40	120	600 / 600	50.0 / 50.0
Dataset B: Stroop + Arithmetic	Rest, Stroop, Arithmetic	40	120	600 / 1,200	33.3 / 66.7
Dataset B: Stroop + Mirror	Rest, Stroop, Mirror	40	120	600 / 1,200	33.3 / 66.7
Dataset B: Arithmetic + Mirror	Rest, Arithmetic, Mirror	40	120	600 / 1,200	33.3 / 66.7

Note. Low and high denote the unified binary attention labels of Table 1. A recording unit is one participant–session–run recording. Counts are the prepared analytical windows, before any training-fold class weighting.

Models [revise need]

We evaluated five classifiers rather than relying on a single decoder to test whether the conclusions are robust to a change in modelling assumptions. This makes the decoder an explicit factor of the design: a difference between validation protocols that is reproduced across models is more credibly attributed to the data and evaluation protocol than to the idiosyncrasies of one learner. The selected models span three major families used in EEG decoding: linear and tree-based traditional machine-learning models, and deep-learning architectures. We place particular emphasis on deep learning because these models learn task-relevant representations directly from the channel-by-time EEG windows, reducing reliance on a manually specified feature set. Hyperparameters were fixed *a priori* for each model and held constant across validation protocols and experimental runs; they were not tuned to an individual protocol or run and are recorded in full in the configuration stored with every run. All models were trained to predict whether a window belonged to a high-demand or a low-demand condition in the sense of Table 1, and not to predict a level of attention.

Traditional machine learning. Logistic regression provides a transparent linear baseline and represents the simplest decision rule supported by the fixed feature representation. XGBoost [26] represents tree-based ensemble learning, allowing nonlinear relations and feature interactions to be modelled within the same representation. It was also the decoder used in the originally submitted version of this study, preserving a direct point of comparison with those results. Both models received a fixed, label-free feature representation. The features were selected to provide complementary and interpretable summaries of each

EEG window: band power describes its spectral distribution across conventional frequency ranges, entropy features characterize signal irregularity and complexity, and descriptive statistics characterize its amplitude distribution. The same prespecified feature set was used in every scenario rather than being selected or tuned separately for a dataset or validation protocol. It comprised log band power in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and gamma (30–45 Hz), together with spectral, approximate, sample and singular-value-decomposition entropy, mean, standard deviation, variance, skewness and kurtosis, computed for each channel. This produced 168 features per window (12 channels $\times$ 14 features per channel). Feature extraction contained no trainable transformation. Transformations that learn from data were fitted within each training fold only: logistic-regression features were standardized on the training data, whereas XGBoost received unscaled features.

Deep learning. The remaining three classifiers operate directly on the prepared channel-by-time EEG windows and learn their representations during training. *ShallowFBCSPNet* [27] is a shallow convolutional neural network designed to emulate the main operations of the filter-bank common spatial pattern (FBCSP) pipeline within a trainable model. It first applies temporal convolution and spatial filtering across channels, then uses a squaring nonlinearity, mean pooling and a logarithmic activation to form band-power-like features. Thus, its inductive bias reflects a conventional EEG decoding approach, while its filters are optimized jointly with the classifier.

EEGNet [28] is a compact convolutional neural network designed specifically for EEG. Its temporal convolutions learn filters over time, depthwise convolutions learn spatial filters across electrodes for each temporal feature map, and separable convolutions combine these learned features efficiently. This factorized design substantially limits the number of trainable parameters while retaining separate temporal and spatial feature learning, making EEGNet the smallest network in the present comparison.

EEG Conformer [29] combines a convolutional front end with a Transformer encoder. The convolutional module forms local temporal and spatial features and converts them into a sequence of embeddings; self-attention in the Transformer then models global dependencies between these embeddings. It therefore complements the local receptive fields of convolution with a mechanism for relating EEG patterns over longer time spans. The only fixed input transformation for the deep-learning models was conversion from volts to microvolts. The networks were trained from scratch. No pretrained weights were used.

[TODO — Which decoder carries the Results narrative?]

Results and analysis

To characterize zero-shot generalization across the prespecified validation strategies, we evaluated the binary contrast between high-demand conditions and rest or task disengagement with five decoders. Table 3 reports decoder-specific performance across these strategies. It presents participant-level balanced accuracy, macro F1, MCC, ROC-AUC, and per-class recall for logistic regression, the interpretable reference decoder. Corresponding tables for XGBoost, EEGNet, ShallowFBCSPNet, and EEGConformer are provided in the Supporting Information.

Table 3. Performance metrics across experimental scenarios for logistic regression.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.64	0.63	0.30	0.73	0.62/0.66
	Stratified K-fold(Full B)	0.68	0.64	0.34	0.74	0.65/0.71
	Stratified K-fold(B: Stroop)	0.69	0.68	0.40	0.74	0.67/0.71
	Stratified K-fold(B: Arithmetic)	0.66	0.65	0.35	0.74	0.66/0.67
	Stratified K-fold(B: Mirror)	0.69	0.67	0.39	0.75	0.66/0.72
Cross-subject	Leave-one-subject-out(Full A)	0.63	0.61	0.29	0.73	0.63/0.63
	Leave-one-subject-out(Full B)	0.66	0.61	0.31	0.74	0.63/0.69
Cross-task (Dataset B)	Stroop → Arithmetic	0.59	0.57	0.21	0.65	0.65/0.54
	Stroop → Mirror	0.66	0.64	0.35	0.73	0.67/0.66
	Arithmetic → Stroop	0.56	0.53	0.14	0.61	0.67/0.46
	Arithmetic → Mirror	0.61	0.56	0.22	0.66	0.65/0.56
	Mirror → Stroop	0.62	0.59	0.27	0.70	0.63/0.62
	Mirror → Arithmetic	0.60	0.57	0.21	0.65	0.66/0.54
	Arithmetic+Mirror → Stroop	0.61	0.56	0.22	0.67	0.65/0.56
	Stroop+Mirror → Arithmetic	0.66	0.64	0.35	0.74	0.59/0.74
	Stroop+Arithmetic → Mirror	0.62	0.60	0.27	0.68	0.62/0.63
	Stroop → Arithmetic+Mirror	0.58	0.55	0.16	0.60	0.65/0.50
	Arithmetic → Stroop+Mirror	0.60	0.56	0.21	0.63	0.63/0.58
	Mirror → Stroop+Arithmetic	0.63	0.60	0.27	0.68	0.64/0.62
Cross-dataset	Full A → Full B	0.52	0.37	0.03	0.54	0.78/0.26
	Full B → Full A	0.55	0.53	0.12	0.58	0.35/0.76
	Full A → Stroop	0.49	0.42	-0.03	0.51	0.71/0.27
	Stroop → Full A	0.54	0.53	0.08	0.56	0.45/0.63
	Full A → Arithmetic	0.53	0.48	0.08	0.58	0.68/0.38
	Arithmetic → Full A	0.52	0.52	0.04	0.54	0.43/0.61
	Full A → Mirror	0.53	0.47	0.06	0.55	0.72/0.34
	Mirror → Full A	0.52	0.48	0.06	0.54	0.26/0.79
	Full A → Stroop+Arithmetic	0.51	0.41	0.02	0.54	0.72/0.30
	Stroop+Arithmetic → Full A	0.52	0.50	0.04	0.53	0.36/0.68
	Full A → Stroop+Mirror	0.51	0.40	0.00	0.52	0.75/0.27
	Stroop+Mirror → Full A	0.53	0.49	0.07	0.54	0.28/0.77
	Full A → Arithmetic+Mirror	0.54	0.43	0.07	0.56	0.77/0.30
	Arithmetic+Mirror → Full A	0.55	0.51	0.11	0.58	0.28/0.81

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve.

Generalization under increasingly stringent validation strategies

To examine generalization under increasingly stringent validation strategies, we compared directed cross-task and cross-dataset transfer with the within-composition baseline of each target composition. These comparisons test whether the binary contrast was retained when training and test data differed in task composition or dataset.

		Target (test)						
Source (train)		Stroop	Arithmetic	Mirror	Full A	Full B	Cross-task mean Δ	Cross-dataset mean Δ
Stroop	Logistic reg	69.3	57.7	62.7	54.0	—	-10.1	-13.0
	XGBoost	69.0	55.7	61.9	50.6	—		
	EEGNet	66.4	55.8	59.2	50.3	—		
	ShallowNet	78.2	59.9	63.9	55.5	—		
	EEGConformer	71.5	55.2	62.8	52.2	—		
Arithmetic	Logistic reg	56.0	66.4	66.2	52.2	—	-7.2	-10.2
	XGBoost	58.2	69.5	66.6	58.9	—		
	EEGNet	59.1	64.5	65.3	50.3	—		
	ShallowNet	62.3	75.1	69.1	55.5	—		
	EEGConformer	60.1	72.1	67.8	59.5	—		
Mirror	Logistic reg	59.7	62.3	68.6	52.4	—	-6.9	-12.9
	XGBoost	61.5	64.3	70.5	53.2	—		
	EEGNet	61.3	62.3	67.6	48.8	—		
	ShallowNet	64.0	68.2	72.9	54.9	—		
	EEGConformer	62.3	67.3	68.6	53.6	—		
Full A	Logistic reg	48.8	53.4	53.2	64.0	52.1	—	-20.6
	XGBoost	47.8	54.0	51.4	67.0	51.6		
	EEGNet	44.7	49.2	45.7	66.0	46.7		
	ShallowNet	46.6	53.2	48.1	65.9	48.8		
	EEGConformer	42.2	52.4	47.3	64.6	47.0		
Full B	Logistic reg	—	—	—	55.2	68.3	—	-10.9
	XGBoost	—	—	—	54.1	68.0		
	EEGNet	—	—	—	51.0	67.1		
	ShallowNet	—	—	—	54.7	72.0		
	EEGConformer	—	—	—	58.2	69.9		

0 25 50 75 100

Fig 1. Zero-shot transfer performance across source and target compositions. Each cell reports participant-level balanced accuracy (%) for one decoder trained on the source composition (rows) and evaluated on the target composition (columns). Grey diagonal cells show the corresponding within-composition baselines; coloured cells show cross-task or cross-dataset transfer; dashes indicate unavailable source–target combinations. For each compatible target composition, each summary first averages participant-level balanced accuracy across the five decoders and subtracts the equally averaged target-specific baseline; it then averages these differences across targets. Negative values indicate lower transfer performance. Cross-task differences combine task-composition and held-out-participant shifts relative to the within-domain baseline.

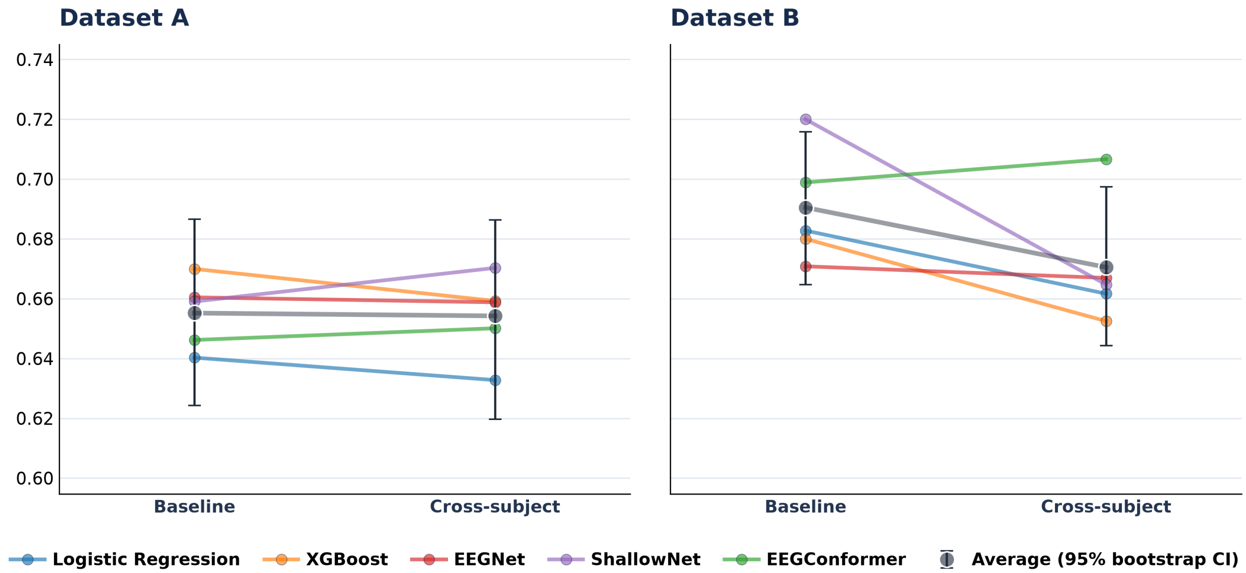

Fig 2. Baseline and cross-subject performance across decoders. For Full A and Full B, coloured lines connect the participant-level mean balanced accuracy of each decoder between baseline and cross-subject validation. Black points show the mean across the five decoders, and bars show paired 95% bootstrap intervals from 10,000 target-participant resamples.

Each directed cross-task estimate was compared with the within-composition reference performance of its target composition. The mean balanced-accuracy degradation across the two other single-task targets was −10.1 percentage points for a Stroop source, −7.2 points for an Arithmetic source, and −6.9 points for a Mirror source (Figure 1). The direction-level means therefore differed between source–target pairs, while each source showed a lower average than its corresponding target-specific baselines. These cross-task results combine a change in task composition with zero-shot generalization to a held-out participant; they do not isolate the effect of task composition alone.

The same target-specific comparison was applied to directed cross-dataset transfers. When Full A was the source, its mean balanced accuracy across the Full B and single-task Dataset B targets was 0.49 (range, 0.46–0.52), a mean degradation of −20.6 percentage points relative to the relevant target baselines. In the reverse Full B→Full A direction, mean balanced accuracy was 0.55 and the degradation was −10.9 points (Figure 1). The directed transfers from the individual Dataset B task compositions to Full A also varied, with mean degradations from −10.2 to −13.0 points. Thus, the observed cross-dataset performance depended on transfer direction under the compound dataset shift represented by this protocol.

Performance consistency across decoders

To assess whether performance patterns across validation strategies depended on decoder choice, we visualized participant-level balanced accuracy for all five decoders. This enabled us to compare the consistency of the observed patterns across decoder families.

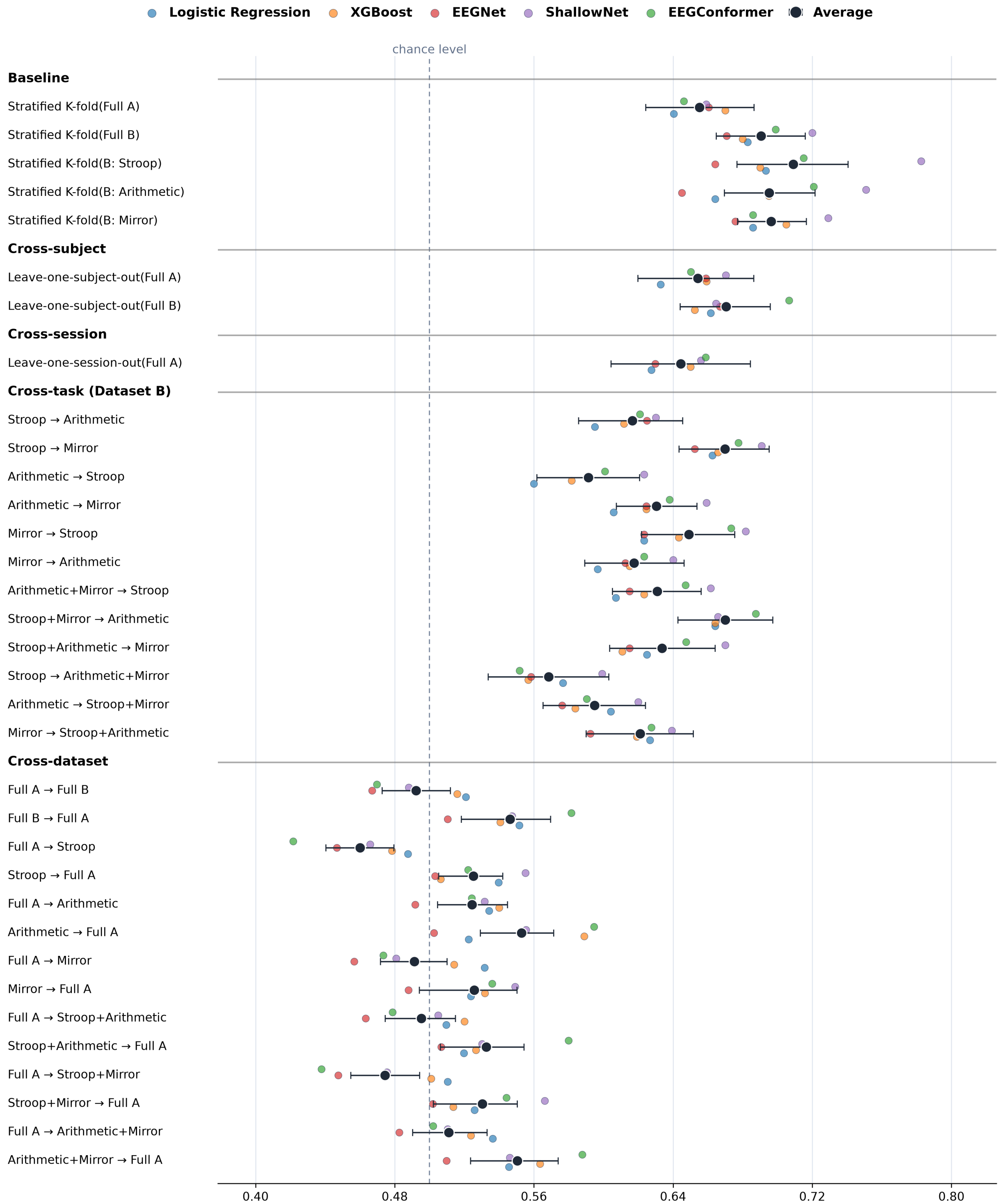

Fig 3. Participant-level balanced accuracy across experimental scenarios for the five decoders. For each decoder and scenario, balanced accuracy was calculated separately for each target participant and averaged across participants (coloured points). Black points show the mean of the five decoder-specific estimates; horizontal bars show paired 95% percentile bootstrap intervals from 10,000 target-participant resamples. The dashed vertical line marks chance-level balanced accuracy (0.50).

Within-composition reference performance was used to evaluate zero-shot generalization to a held-out target participant. Averaged across the five decoders, balanced accuracy was 0.66 for the Full A baseline and 0.69 for the Full B baseline (Figure 3). When the target participant was held out, the corresponding all-decoder means were 0.65 for Full A and 0.67 for Full B. Thus, the discrimination of the protocol-defined high-demand versus rest or task-disengagement contrast was retained under the participant shifts evaluated here, with the pattern reproduced across the five decoders.

Discussion

[**TODO — Discussion paragraph 1: Summary of the paper's main conclusion.**] Lorem ipsum dolor sit amet, consectetur adipiscing elit.

[**TODO — Discussion paragraph 2: Comparison with previous results or theories.**] Lorem ipsum dolor sit amet, consectetur adipiscing elit.

[**TODO — Discussion paragraph 3: Scientific or engineering implications of this work.**] Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Several limitations delimit the interpretation of these findings. First, the binary labels represent protocol-defined contrasts and do not isolate attention from related differences in arousal, visual input, stress, or task demands. Second, restriction to the 12 channels shared by the datasets limits the stability of ICA-based artefact removal. Third, the common preparation pipeline was deliberately parsimonious and fixed across datasets, task compositions, validation protocols, and decoders. The reported performance should therefore not be interpreted as the maximum attainable for any individual scenario: more extensive, dataset-specific data cleaning and preparation could alter performance, but evaluating such optimization was outside the aim of maintaining a common analytical basis for comparative zero-shot evaluation. Finally, all source–target transfers were evaluated without target-specific domain adaptation or personalization. The present results thus provide pre-adaptation baselines for source–target suitability, but do not quantify the benefit of adaptation relative to the corresponding zero-shot condition. Future work should evaluate these factors explicitly while retaining leakage-resistant splits and clearly separating improvements in optimized within-domain performance from gains in cross-domain generalization.

[**TODO — Discussion paragraph 5: Forward-looking statement.**] Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Conclusion

[TODO — Conclusion section is currently in progress.]

[TODO — Add acknowledgments if applicable.]

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Supporting information

[TODO — Supporting information is currently in progress.]

Table 4. Performance metrics across experimental scenarios for XGBoost.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.67	0.67	0.35	0.74	0.64/0.70
	Stratified K-fold(Full B)	0.68	0.65	0.36	0.75	0.60/0.76
	Stratified K-fold(B: Stroop)	0.69	0.68	0.40	0.76	0.68/0.70
	Stratified K-fold(B: Arithmetic)	0.70	0.68	0.41	0.76	0.69/0.70
	Stratified K-fold(B: Mirror)	0.70	0.70	0.43	0.77	0.72/0.69
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.65	0.33	0.74	0.65/0.67
	Leave-one-subject-out(Full B)	0.65	0.61	0.31	0.73	0.58/0.72
Cross-task (Dataset B)	Stroop → Arithmetic	0.61	0.59	0.24	0.65	0.64/0.58
	Stroop → Mirror	0.67	0.64	0.36	0.74	0.66/0.67
	Arithmetic → Stroop	0.58	0.55	0.18	0.62	0.68/0.48
	Arithmetic → Mirror	0.62	0.58	0.25	0.69	0.67/0.58
	Mirror → Stroop	0.64	0.62	0.32	0.72	0.70/0.59
	Mirror → Arithmetic	0.61	0.58	0.25	0.67	0.72/0.51
	Arithmetic+Mirror → Stroop	0.62	0.57	0.25	0.70	0.70/0.55
	Stroop+Mirror → Arithmetic	0.66	0.64	0.35	0.74	0.60/0.72
	Stroop+Arithmetic → Mirror	0.61	0.58	0.24	0.67	0.63/0.59
	Stroop → Arithmetic+Mirror	0.56	0.52	0.13	0.58	0.67/0.45
	Arithmetic → Stroop+Mirror	0.58	0.53	0.18	0.63	0.63/0.53
	Mirror → Stroop+Arithmetic	0.62	0.59	0.27	0.69	0.65/0.59
Cross-dataset	Full A → Full B	0.52	0.49	0.04	0.52	0.28/0.75
	Full B → Full A	0.54	0.52	0.09	0.55	0.37/0.71
	Full A → Stroop	0.48	0.42	-0.04	0.47	0.26/0.70
	Stroop → Full A	0.51	0.48	0.02	0.50	0.32/0.69
	Full A → Arithmetic	0.54	0.47	0.10	0.55	0.27/0.81
	Arithmetic → Full A	0.59	0.58	0.18	0.62	0.63/0.55
	Full A → Mirror	0.51	0.45	0.05	0.52	0.27/0.76
	Mirror → Full A	0.53	0.52	0.07	0.56	0.61/0.45
	Full A → Stroop+Arithmetic	0.52	0.49	0.05	0.51	0.29/0.75
	Stroop+Arithmetic → Full A	0.53	0.51	0.06	0.53	0.37/0.68
	Full A → Stroop+Mirror	0.50	0.47	0.01	0.50	0.27/0.73
	Stroop+Mirror → Full A	0.51	0.49	0.03	0.52	0.38/0.65
	Full A → Arithmetic+Mirror	0.52	0.49	0.07	0.54	0.27/0.78
	Arithmetic+Mirror → Full A	0.56	0.56	0.13	0.58	0.57/0.56

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve.

Table 5. Performance metrics across experimental scenarios for EEGNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.65	0.34	0.75	0.56/0.76
	Stratified K-fold(Full B)	0.67	0.64	0.34	0.74	0.57/0.77
	Stratified K-fold(B: Stroop)	0.66	0.65	0.36	0.71	0.58/0.74
	Stratified K-fold(B: Arithmetic)	0.64	0.63	0.31	0.72	0.60/0.69
	Stratified K-fold(B: Mirror)	0.68	0.67	0.37	0.74	0.60/0.75
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.63	0.35	0.78	0.66/0.66
	Leave-one-subject-out(Full B)	0.67	0.62	0.33	0.75	0.60/0.74
Cross-task (Dataset B)	Stroop → Arithmetic	0.62	0.59	0.27	0.70	0.59/0.66
	Stroop → Mirror	0.65	0.62	0.34	0.73	0.60/0.71
	Arithmetic → Stroop	0.59	0.55	0.21	0.66	0.61/0.57
	Arithmetic → Mirror	0.62	0.58	0.28	0.69	0.61/0.64
	Mirror → Stroop	0.62	0.60	0.28	0.69	0.57/0.68
	Mirror → Arithmetic	0.61	0.58	0.24	0.68	0.57/0.65
	Arithmetic+Mirror → Stroop	0.61	0.58	0.25	0.68	0.57/0.66
	Stroop+Mirror → Arithmetic	0.67	0.65	0.37	0.75	0.61/0.73
	Stroop+Arithmetic → Mirror	0.61	0.59	0.26	0.69	0.60/0.63
	Stroop → Arithmetic+Mirror	0.56	0.52	0.13	0.59	0.65/0.47
	Arithmetic → Stroop+Mirror	0.58	0.52	0.16	0.63	0.59/0.56
	Mirror → Stroop+Arithmetic	0.59	0.56	0.21	0.67	0.60/0.58
Cross-dataset	Full A → Full B	0.47	0.34	-0.07	0.46	0.61/0.32
	Full B → Full A	0.51	0.48	0.04	0.53	0.30/0.72
	Full A → Stroop	0.45	0.37	-0.13	0.39	0.60/0.29
	Stroop → Full A	0.50	0.46	0.01	0.50	0.27/0.74
	Full A → Arithmetic	0.49	0.43	-0.00	0.52	0.58/0.40
	Arithmetic → Full A	0.50	0.43	0.01	0.51	0.16/0.84
	Full A → Mirror	0.46	0.38	-0.10	0.46	0.60/0.32
	Mirror → Full A	0.49	0.48	-0.02	0.48	0.44/0.53
	Full A → Stroop+Arithmetic	0.46	0.37	-0.08	0.46	0.60/0.32
	Stroop+Arithmetic → Full A	0.51	0.39	0.03	0.57	0.08/0.94
	Full A → Stroop+Mirror	0.45	0.35	-0.12	0.43	0.60/0.29
	Stroop+Mirror → Full A	0.50	0.39	-0.00	0.49	0.08/0.93
	Full A → Arithmetic+Mirror	0.48	0.39	-0.04	0.49	0.60/0.37
	Arithmetic+Mirror → Full A	0.51	0.47	0.03	0.51	0.28/0.74

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve.

Table 6. Performance metrics across experimental scenarios for ShallowFBCSPNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.65	0.33	0.74	0.63/0.69
	Stratified K-fold(Full B)	0.72	0.72	0.46	0.81	0.58/0.86
	Stratified K-fold(B: Stroop)	0.78	0.78	0.58	0.86	0.77/0.79
	Stratified K-fold(B: Arithmetic)	0.75	0.75	0.52	0.83	0.76/0.74
	Stratified K-fold(B: Mirror)	0.73	0.72	0.47	0.80	0.72/0.74
Cross-subject	Leave-one-subject-out(Full A)	0.67	0.66	0.37	0.76	0.72/0.62
	Leave-one-subject-out(Full B)	0.66	0.64	0.36	0.80	0.52/0.81
Cross-task (Dataset B)	Stroop → Arithmetic	0.63	0.59	0.28	0.72	0.63/0.63
	Stroop → Mirror	0.69	0.67	0.41	0.79	0.73/0.65
	Arithmetic → Stroop	0.62	0.59	0.27	0.72	0.72/0.53
	Arithmetic → Mirror	0.66	0.61	0.33	0.76	0.73/0.59
	Mirror → Stroop	0.68	0.66	0.40	0.77	0.69/0.68
	Mirror → Arithmetic	0.64	0.62	0.31	0.71	0.72/0.56
	Arithmetic+Mirror → Stroop	0.66	0.62	0.33	0.74	0.70/0.62
	Stroop+Mirror → Arithmetic	0.67	0.64	0.36	0.78	0.61/0.72
	Stroop+Arithmetic → Mirror	0.67	0.64	0.37	0.76	0.62/0.72
	Stroop → Arithmetic+Mirror	0.60	0.56	0.22	0.68	0.72/0.48
	Arithmetic → Stroop+Mirror	0.62	0.56	0.25	0.71	0.70/0.54
	Mirror → Stroop+Arithmetic	0.64	0.61	0.30	0.73	0.73/0.55
Cross-dataset	Full A → Full B	0.49	0.29	-0.04	0.48	0.79/0.19
	Full B → Full A	0.55	0.48	0.10	0.58	0.28/0.81
	Full A → Stroop	0.47	0.37	-0.10	0.39	0.77/0.16
	Stroop → Full A	0.56	0.52	0.12	0.60	0.52/0.59
	Full A → Arithmetic	0.53	0.45	0.07	0.58	0.74/0.32
	Arithmetic → Full A	0.56	0.52	0.14	0.61	0.55/0.56
	Full A → Mirror	0.48	0.39	-0.06	0.49	0.78/0.19
	Mirror → Full A	0.55	0.53	0.11	0.58	0.53/0.57
	Full A → Stroop+Arithmetic	0.51	0.35	-0.00	0.49	0.80/0.21
	Stroop+Arithmetic → Full A	0.53	0.48	0.07	0.57	0.38/0.68
	Full A → Stroop+Mirror	0.48	0.32	-0.08	0.44	0.78/0.17
	Stroop+Mirror → Full A	0.57	0.54	0.14	0.60	0.60/0.53
	Full A → Arithmetic+Mirror	0.51	0.37	0.01	0.53	0.78/0.24
	Arithmetic+Mirror → Full A	0.55	0.53	0.11	0.59	0.65/0.45

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve.

Table 7. Performance metrics across experimental scenarios for EEGConformer.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.65	0.64	0.31	0.73	0.70/0.59
	Stratified K-fold(Full B)	0.70	0.69	0.41	0.80	0.55/0.84
	Stratified K-fold(B: Stroop)	0.72	0.70	0.46	0.79	0.59/0.84
	Stratified K-fold(B: Arithmetic)	0.72	0.71	0.47	0.80	0.61/0.83
	Stratified K-fold(B: Mirror)	0.69	0.67	0.40	0.80	0.59/0.78
Cross-subject	Leave-one-subject-out(Full A)	0.65	0.64	0.32	0.76	0.66/0.64
	Leave-one-subject-out(Full B)	0.71	0.68	0.42	0.80	0.60/0.82
Cross-task (Dataset B)	Stroop → Arithmetic	0.62	0.59	0.25	0.70	0.65/0.59
	Stroop → Mirror	0.68	0.65	0.38	0.78	0.62/0.73
	Arithmetic → Stroop	0.60	0.56	0.22	0.68	0.61/0.59
	Arithmetic → Mirror	0.64	0.59	0.28	0.73	0.62/0.66
	Mirror → Stroop	0.67	0.65	0.39	0.77	0.64/0.71
	Mirror → Arithmetic	0.62	0.60	0.27	0.68	0.65/0.60
	Arithmetic+Mirror → Stroop	0.65	0.61	0.30	0.73	0.65/0.65
	Stroop+Mirror → Arithmetic	0.69	0.66	0.41	0.79	0.56/0.81
	Stroop+Arithmetic → Mirror	0.65	0.61	0.33	0.74	0.61/0.69
	Stroop → Arithmetic+Mirror	0.55	0.51	0.12	0.62	0.67/0.44
	Arithmetic → Stroop+Mirror	0.59	0.53	0.20	0.67	0.62/0.56
	Mirror → Stroop+Arithmetic	0.63	0.60	0.29	0.71	0.63/0.63
Cross-dataset	Full A → Full B	0.47	0.31	-0.07	0.43	0.70/0.24
	Full B → Full A	0.58	0.56	0.18	0.64	0.62/0.54
	Full A → Stroop	0.42	0.34	-0.18	0.33	0.65/0.20
	Stroop → Full A	0.52	0.43	0.07	0.62	0.13/0.91
	Full A → Arithmetic	0.52	0.45	0.05	0.56	0.60/0.45
	Arithmetic → Full A	0.59	0.57	0.20	0.66	0.68/0.50
	Full A → Mirror	0.47	0.39	-0.05	0.42	0.66/0.29
	Mirror → Full A	0.54	0.50	0.08	0.57	0.76/0.31
	Full A → Stroop+Arithmetic	0.48	0.36	-0.05	0.44	0.68/0.27
	Stroop+Arithmetic → Full A	0.58	0.55	0.18	0.64	0.58/0.58
	Full A → Stroop+Mirror	0.44	0.32	-0.14	0.37	0.65/0.23
	Stroop+Mirror → Full A	0.54	0.49	0.11	0.61	0.31/0.78
	Full A → Arithmetic+Mirror	0.50	0.39	-0.00	0.49	0.68/0.32
	Arithmetic+Mirror → Full A	0.59	0.57	0.19	0.63	0.60/0.57

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve.